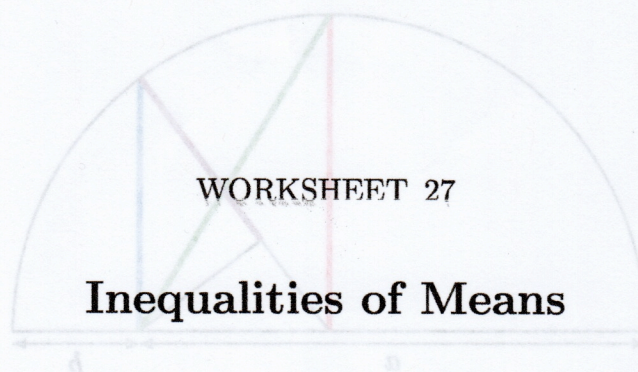




WORKSHEET 27

Inequalities of Means

Given two positive numbers a and b , their *arithmetic mean* is $\frac{a+b}{2}$, whereas their *geometric mean* is $\sqrt{ab}$.

PROBLEM 27.1. Show that for any two positive numbers a and b , the arithmetic mean of a and b is greater than or equal to their geometric mean, i.e. $\frac{a+b}{2} \geq \sqrt{ab}$. Under what condition do we get equality?

This result is (a simple case of) one of the most important inequalities, called the *AM-GM inequality*.

PROBLEM 27.2. Show that if $a > 0$, then $a + \frac{1}{a} \geq 2$.

PROBLEM 27.3. Show that if a, b, c are positive, then $a^2 + b^2 + c^2 \geq ab + bc + ca$.

PROBLEM 27.4. Show that if a, b, c are positive, then $(a+b)(b+c)(c+a) \geq 8abc$.

The arithmetic mean and the geometric mean are the two most important types of means in mathematics, but there are some others that occasionally show up. The next two most important are called the *harmonic mean* and the *quadratic mean* (also known as the *root mean square*):

$$HM = \frac{2}{\frac{1}{a} + \frac{1}{b}}, \quad QM = \sqrt{\frac{a^2 + b^2}{2}}.$$

PROBLEM 27.5. Show that for any positive numbers a and b , we have

$$QM \geq AM \geq GM \geq HM.$$

PROBLEM 27.6. Explain the pictorial proof of these inequalities shown in Figure 1.

These inequalities also hold for more numbers, if we generalize the means in the right way. If we have n positive numbers $a_1, \dots, a_n$, then

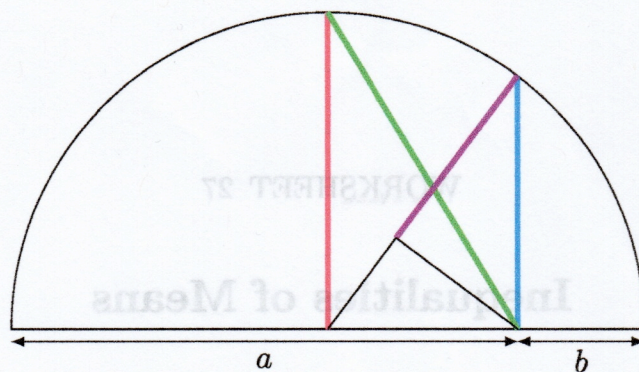


Figure 1. A pictorial proof of the inequalities $QM \geq AM \geq GM \geq HM$.

we define

$$QM = \sqrt{\frac{a_1^2 + a_2^2 + \cdots + a_n^2}{n}},$$

$$AM = \frac{a_1 + a_2 + \cdots + a_n}{n},$$

$$GM = \sqrt[n]{a_1 a_2 \cdots a_n},$$

$$HM = \frac{1}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}}.$$

PROBLEM 27.7. Show that the AM-GM inequality holds for $n = 4$ by using the AM-GM inequality for $n = 2$ several times.

PROBLEM 27.8. Deduce the AM-GM inequality for $n = 3$ from the AM-GM inequality for $n = 4$ by making a clever choice for the fourth variable.

PROBLEM 27.9. Show that the AM-GM inequality holds when n is a power of 2.

PROBLEM 27.10. Show that the AM-GM inequality holds for any number of variables.

PROBLEM 27.11. Show that $GM \geq HM$ for any number of variables.

PROBLEM 27.12. Show that $QM \geq AM$ for any number of variables.

There is a great generalization of all these inequalities, called the *power mean inequality*. Given positive numbers $a_1, \dots, a_n$ and a nonzero real number k , we define the k^{th} power mean to be

$$P_k(a_1, \dots, a_n) = \left(\frac{a_1^k + a_2^k + \cdots + a_n^k}{n} \right)^{1/k}.$$

same idea,
more values

$$\frac{a+b+c+d}{2} = \sqrt{ab} + \sqrt{cd} \quad \frac{(a+b+c+d)^2}{4} = ab + cd + 2\sqrt{abcd}$$

$$\frac{(a+b+c+d)^2}{4} = \frac{ab+cd}{2} + \sqrt{abcd}$$

27. INEQUALITIES OF MEANS

The power mean inequality says that if $m \geq k$, then

$$P_m(a_1, \dots, a_n) \geq P_k(a_1, \dots, a_n).$$

Since the quadratic, arithmetic, and harmonic means are the power means for $k = 2$, $k = 1$, and $k = -1$, respectively, this implies the inequalities among their respective means. As for the geometric means, note that the 0^{th} power mean is not defined, but we can still take the k^{th} power mean for k close to 0. It turns out that the geometric mean is the limit of the k^{th} power means as k tends to 0. However, the proof of these facts is a bit beyond the scope of this book, as it relies on more advanced inequalities (in particular, Jensen's inequality).

$$a^2 + b^2 + c^2 + d^2 + 2ab + 2ac + 2ad + 2bc + 2bd + 2cd$$

$$= \frac{(a+b+c+d)^2}{1} = \frac{a^2 + b^2 + c^2 + d^2 + 2ab + 2ac + 2ad + 2bc + 2bd + 2cd}{1}$$

$$= \frac{a^2 + b^2 + c^2 + d^2 + 2ab + 2ac + 2ad + 2bc + 2bd + 2cd}{1}$$

Inequalities of Means *Turns out
Alexander Friesen 5/12-17/26 I didn't have to.

27.1: Assume the contrary*, where $\frac{1}{2}(a+b) < \sqrt{ab}$.

Square both sides: $\frac{1}{4}(a+b)^2 = \frac{1}{4}(a^2+b^2+2ab) < ab$.

$$a^2+b^2+2ab < 4ab, a^2+b^2-2ab < 0,$$

Factor: $(a-b)^2 < 0 \rightarrow$ impossible - squares
are always nonnegative! Therefore,

it is always true that $\frac{1}{2}(a+b) \geq \sqrt{ab}$.

27.2: Using 27.1's logic:

$$(a-1)^2 \geq 0$$

$$a^2-2a+1 \geq 0$$

$$a^2+1 \geq 2a, \text{ Divide by } a:$$

$$\boxed{a + \frac{1}{a} \geq 2.}$$

27.3: Same logic, with 3 variables:

$$(a-b)^2 \geq 0, (b-c)^2 \geq 0, (c-a)^2 \geq 0.$$

Added together they are ≥ 0 . Expand:

$$a^2-2ab+b^2+b^2-2bc+c^2+c^2-2ca+a^2 \geq 0$$

$$2a^2+2b^2+2c^2-2ab-2bc-2ca \geq 0$$

$$2a^2+2b^2+2c^2 \geq 2ab+2bc+2ca$$

$$\boxed{a^2+b^2+c^2 \geq ab+bc+ca}$$

27.4: If $a+b \geq 2\sqrt{ab}$, $b+c \geq 2\sqrt{bc}$, and $c+a \geq 2\sqrt{ca}$,

$$\text{then } \boxed{(a+b)(b+c)(c+a) \geq 2\sqrt{ab} \cdot 2\sqrt{bc} \cdot 2\sqrt{ca} = 8\sqrt{a^2b^2c^2} = 8\sqrt{abc}.$$

27.5: Two proofs.

QM $\geq$ AM: Setup: $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$.

Steps Alg: $\frac{a^2+b^2}{2} \geq \frac{a^2+b^2+2ab}{4}$

$$2(a^2+b^2) \geq a^2+b^2+2ab$$

$$a^2+b^2 \geq 2ab$$

$$a^2+b^2-2ab \geq 0$$

$(a-b)^2 \geq 0$. Must be true,

so QM $\geq$ AM because a square is nonnegative.

HM $\leq$ GM: Setup: $\frac{2}{\frac{1}{a}+\frac{1}{b}} \leq \sqrt{ab}$

Steps Alg: $\frac{1}{\frac{1}{a}+\frac{1}{b}} \geq \frac{1}{\frac{1}{\sqrt{ab}}} = \frac{\sqrt{ab}}{ab}$

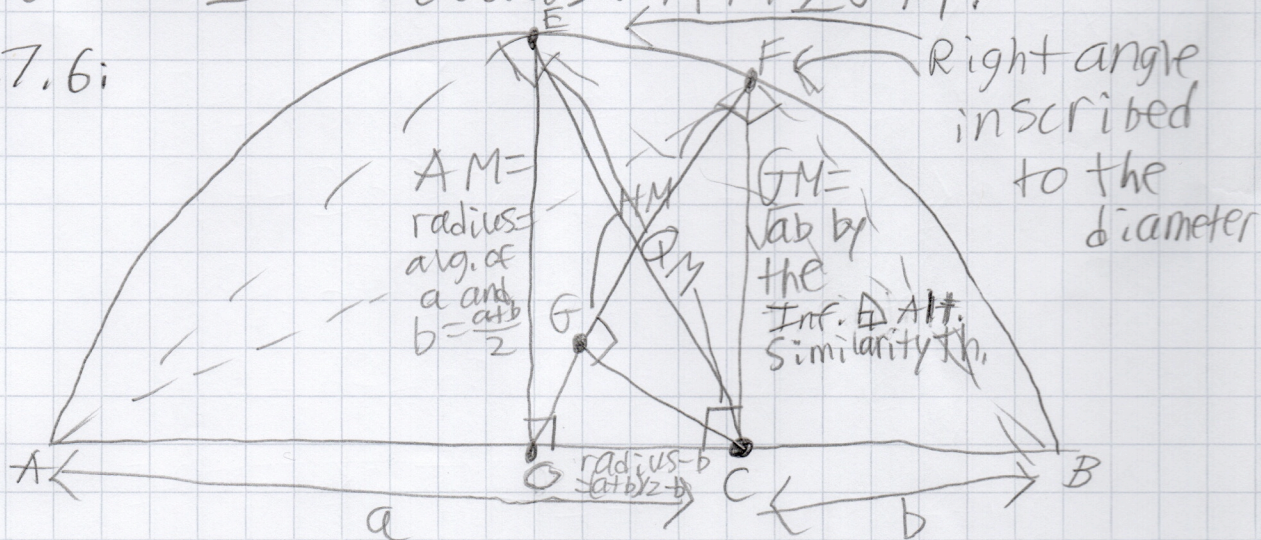
$$\frac{1}{\frac{1}{a}+\frac{1}{b}} \geq \frac{2}{\frac{1}{\sqrt{ab}}} = \frac{2\sqrt{ab}}{ab}$$

$$b+a \geq 2\sqrt{ab}$$

$\frac{a+b}{2} \geq \sqrt{ab}$. Must be true by 27.1,

so HM $\leq$ GM because AM $\geq$ GM.

27.6:



QM = EC = Pythagoras of $\frac{a+b}{2}$ and $(\frac{a+b}{2} - b) = \sqrt{\frac{a^2+b^2}{2}}$
 FG = HM by similar Δ s FGC and FCE ($\frac{HM}{GM} = \frac{FM}{OF}$) GM and OF are known.

27.7: Suppose we wish to prove $\frac{a+b+c+d}{4} = \sqrt[4]{abcd}$.

We already know $\frac{a+b}{2} \geq \sqrt{ab}$, $\frac{b+c}{2} \geq \sqrt{bc}$, $\frac{c+d}{2} \geq \sqrt{cd}$.

Take the average^(Both) of $\frac{a+b}{2} \geq \sqrt{ab}$ and $\frac{c+d}{2} \geq \sqrt{cd}$:

$$\frac{\frac{a+b}{2} + \frac{c+d}{2}}{2} \geq \sqrt{\frac{a+b}{2} \cdot \frac{c+d}{2}} \geq \sqrt{\sqrt{ab} \cdot \sqrt{cd}} = \sqrt[4]{abcd}.$$

Simplifying out middle term:

$$\boxed{\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}}. \text{ AM-GM holds at 4.}$$

27.8: $\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$. Let $d = \sqrt[3]{abc}$:

$$\sqrt[4]{abcd} = \sqrt[4]{abc \cdot \sqrt[3]{abc}} = \sqrt[4]{abc \cdot (abc)^{\frac{1}{3}}} = \sqrt[4]{(abc)^{\frac{4}{3}}} = \sqrt[3]{abc}.$$

$$\frac{a+b+c+d}{4} \geq \sqrt[3]{abc}, \quad \frac{a+b+c+\sqrt[3]{abc}}{4} \geq \sqrt[3]{abc},$$

$$a+b+c+\sqrt[3]{abc} \geq 4\sqrt[3]{abc}, \quad a+b+c \geq 3\sqrt[3]{abc},$$

$$\text{Final eq: } \boxed{\frac{a+b+c}{3} \geq \sqrt[3]{abc}}.$$

27.9: Induction proof: If AM-GM for n values, it

should also work for $2n$ values.

Call these values $x_1, x_2, \dots, x_{n-1}, x_{2n}$.

If n values work, then $\frac{x_1+x_2+\dots+x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}$ and

likewise for $x_{n+1}, x_{n+2}, \dots, x_{2n-1}, x_{2n}$.

Taking the average of these 2 AM-GM's:

$$\frac{\frac{x_1+x_2+\dots+x_n}{n} + \frac{x_{n+1}+x_{n+2}+\dots+x_{2n}}{n}}{2} \geq \sqrt[2]{\sqrt[n]{x_1 x_2 \dots x_n} \cdot \sqrt[n]{x_{n+1} x_{n+2} \dots x_{2n}}}, \text{ which}$$

equals $\frac{x_1+x_2+\dots+x_{2n-1}+x_{2n}}{2n} \geq \sqrt[2n]{x_1 x_2 \dots x_{2n-1} x_{2n}}$. Generalized to 2^n !



$$\frac{A^{2^k}}{A^m} = A^{2^k - m} \geq x_1 x_2 \dots x_n. \text{ But } 2^k - m = n, \text{ so } A^n \geq x_1 x_2 \dots x_n.$$

$$\frac{A^2}{A^m} = A^{2^k - m} \geq x_1 x_2 \dots x_n. \text{ But } 2^k - m = n, \text{ so } A^n \geq x_1 x_2 \dots x_n.$$

27.11: HM $\leq$ GM Setup: $\frac{1}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} \leq \sqrt[n]{x_1 x_2 \dots x_n}$.

Alg Steps: $\frac{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}{n} \geq \frac{1}{\sqrt[n]{x_1 x_2 \dots x_n}}, \frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \geq \frac{n}{\sqrt[n]{x_1 x_2 \dots x_n}}$
 $x_2 x_3 \dots x_n + x_1 x_3 \dots x_n + x_1 x_2 x_4 \dots x_n \dots \geq n \cdot \sqrt[n]{x_1 x_2 \dots x_n} \cdot \sqrt[n]{x_1 x_2 \dots x_n}^{n-1}$
 $\parallel$
 $\geq n \cdot \sqrt[n]{x_1 x_2 \dots x_n}^{n-1}$

So, $HM = \frac{n \cdot (\oplus M)^n}{n} \geq \frac{n \cdot (\oplus M)^n}{n \cdot (\oplus M)^{n-1}} = \oplus M$, and therefore $HM \leq \oplus M$.

27.12: QM $\geq$ AM Setup: $\sqrt{\frac{x_1^2 + x_2^2 + \dots + x_n^2}{n}} \geq \frac{x_1 + x_2 + \dots + x_n}{n}$. Alg: $\frac{x_1^2 + \dots + x_n^2}{n} \geq \frac{(x_1 + \dots + x_n)^2}{n^2}$

$$n(x_1^2 + \dots + x_n^2) \geq (x_1 + \dots + x_n)^2 \quad n x_1^2 + \dots + n x_n^2 \geq (x_1^2 + \dots + x_n^2) + 2(x_1 x_2 + x_1 x_3 + \dots + x_{n-1} x_n)$$

$$(n-1)x_1^2 + (n-1)x_2^2 + \dots + (n-1)x_n^2 - 2(x_1x_2 + \dots + x_{n-1}x_n) \geq 0. \text{ Factor the sum:}$$

Each x^2 appears $n-1$ times, so the expression can factor to $(x_1 - x_2)^2 + (x_2 - x_3)^2 + \dots + (x_{n-1} - x_n)^2 \geq 0$. (something) $^2 \geq 0$, so the QM $\geq$ AM must be true.